

README

HelixCode Video Course Player System

A professional, production-ready video course platform for the HelixCode website. This system provides a complete learning experience with video playback, progress tracking, interactive features, and course management.

File Structure

```
courses/
├── index.html           # Main video player page
├── catalog.html        # Course catalog/listing page
├── course.html         # Individual course detail page
├── player.css          # Custom video player styles
├── player.js           # Video player controller
├── course-data.js      # Course metadata management
├── progress.js         # Progress tracking system
├── search.js           # Search and filtering functionality
└── README.md           # This file
```

Features

Video Player (`index.html` + `player.js`)

Core Playback: - HTML5 video with custom controls - Play/pause, seek, volume control - Playback speed control (0.5x - 2x) - Quality selector (placeholder for future) - Fullscreen support with API integration

Advanced Features: - Interactive transcript synchronized with video - Bookmark functionality with notes - Auto-advance to next chapter - Resume from last position - Chapter navigation with sidebar - Loading states and buffering indicators

Keyboard Shortcuts: - Space - Play/Pause - → - Forward 10 seconds - ← - Backward 10 seconds - ↑ - Volume up - ↓ - Volume down - F - Toggle fullscreen - M - Mute/unmute - C - Toggle subtitles - N - Toggle notes panel - B - Add bookmark - ? - Show shortcuts help

Progress Tracking: - Real-time progress saving to localStorage - Course completion percentage - Chapter completion status - Recently watched tracking - Study streak calculation

Note Taking: - Collapsible notes panel - Auto-save functionality - Character count - Per-course storage

Course Catalog (catalog.html + search.js)

Discovery: - Grid layout of all courses - Course cards with thumbnails and metadata - “Continue Learning” section for resumed courses - Search functionality across titles and descriptions - Filter by difficulty level (Beginner/Intermediate/Advanced) - Sort by newest, popular, level, or duration

Course Cards Display: - Course level badge - Progress indicator for started courses - Duration and chapter count - Description and instructor - Start/Continue button

Course Detail Page (course.html)

Information: - Comprehensive course overview - Learning objectives list - Complete chapter list with descriptions - Prerequisites and requirements - Instructor information - Course includes (duration, resources, certificate)

Preview: - Course preview thumbnail - Progress tracking display - Quick start functionality - Chapter navigation

Progress Tracking System (progress.js)

Tracking Capabilities: - Chapter-level progress (time watched) - Course completion percentage - Last watched timestamp - Completion status - Study streaks

Statistics: - Total watch time - Courses started/completed - Chapters completed - Recent activity - Achievement system

Data Management: - localStorage persistence - Export/import functionality - Progress reset options - Certificate eligibility checking

Achievements: - Getting Started (first course) - Dedicated Learner (5 chapters) - Week Warrior (7-day streak) - Course Master (first completion) - Expert Learner (3 courses)

Course Data Management (course-data.js)

Course Structure:

```

{
  id: 'course-id',
  title: 'Course Title',
  description: 'Course description',
  instructor: 'Instructor Name',
  duration: '2h 30m',
  level: 'beginner|intermediate|advanced',
  chapters: [
    {
      id: 'chapter-id',
      number: 1,
      title: 'Chapter Title',
      description: 'Chapter description',
      duration: '15:00',
      videoUrl: 'video-url',
      transcript: [
        { time: 0, text: 'Transcript text' }
      ],
      resources: [
        { name: 'Resource Name', url: 'resource-url' }
      ]
    }
  ]
}

```

Sample Courses Included: 1. Introduction to HelixCode (Beginner, 2h 30m, 6 chapters) 2. Advanced HelixCode Features (Advanced, 3h 15m, 4 chapters) 3. Production Deployment (Intermediate, 2h 45m, 4 chapters)

Design Features

Responsive Design

- Desktop: Full sidebar and notes panel
- Tablet: Collapsible sidebar
- Mobile: Bottom navigation, vertical layout

Theme Support

- Light theme (default)
- Dark theme (auto-detection via prefers-color-scheme)
- Matches HelixCode brand colors

Accessibility

- ARIA labels on all interactive elements
- Keyboard navigation support
- Focus visible indicators
- Screen reader friendly
- Semantic HTML structure

Performance

- Lazy loading of video content
- Efficient DOM updates
- Debounced search input
- Optimized localStorage usage
- Smooth animations (CSS transitions)

Setup & Usage

Integration with Existing Site

1. **Add to Navigation:** Update main site navigation to include courses link:

```
<li><a href="courses/catalog.html" class="nav-link">Courses</a></li>
```

2. **Video Content:** Currently uses sample videos. Replace with actual course videos:

- Update `videoUrl` in `course-data.js`
- Add your video files to appropriate hosting
- Supported formats: MP4, WebM, OGG

3. **Customization:**

- Colors: Edit CSS variables in `player.css`
- Branding: Update logos and text
- Course content: Modify `course-data.js`

Adding New Courses

1. **Define Course Data:**

```
CourseData.courses['new-course-id'] = {  
  id: 'new-course-id',  
  title: 'New Course Title',  
  description: 'Course description',
```

```

    instructor: 'Instructor Name',
    duration: '3h 00m',
    level: 'intermediate',
    chapters: [/* chapter objects */]
};

```

2. Add Chapter Content:

```

{
  id: 'ch1-new',
  number: 1,
  title: 'Chapter Title',
  description: 'Chapter description',
  duration: '20:00',
  videoUrl: 'path/to/video.mp4',
  transcript: [
    { time: 0, text: 'Transcript text' },
    { time: 15, text: 'More text' }
  ],
  resources: [
    { name: 'Slides (PDF)', url: 'path/to/resource.pdf' }
  ]
}

```

3. Test:

- Load catalog page
- Click on new course
- Verify video playback
- Check progress tracking

Transcript Format

Transcripts are arrays of time-stamped text:

```

transcript: [
  { time: 0, text: 'Welcome to this course.' },
  { time: 15, text: 'In this section, we will cover...' },
  { time: 45, text: 'Let\'s get started with...' }
]

```

Time is in seconds from video start.

Data Storage

localStorage Keys

- helixcode_course_progress - Chapter progress data
- helixcode_bookmarks - Video bookmarks
- helixcode_notes_{courseId} - Course notes

Data Structure

Progress:

```
{
  "chapter-id": {
    started: "2025-01-01T00:00:00.000Z",
    completed: true,
    completedAt: "2025-01-02T00:00:00.000Z",
    currentTime: 1234.56,
    lastWatched: "2025-01-02T00:00:00.000Z"
  }
}
```

Bookmarks:

```
[
  {
    id: 1234567890,
    chapterId: "ch1-intro",
    time: 123.45,
    note: "Important concept",
    timestamp: "2025-01-01T00:00:00.000Z"
  }
]
```

API Reference

VideoPlayer Class

```
// Main player controller
const player = new VideoPlayer();

// Methods
player.loadChapter(chapterId);
player.togglePlay();
```

```
player.skip(seconds);  
player.addBookmark();  
player.toggleFullscreen();  
player.setPlaybackSpeed(speed);
```

ProgressTracker Class

```
// Progress tracking  
const tracker = new ProgressTrackingSystem();  
  
// Methods  
tracker.updateChapterProgress(chapterId, currentTime);  
tracker.markChapterComplete(chapterId);  
tracker.isChapterComplete(chapterId);  
tracker.getCourseProgress(courseId);  
tracker.getStats();  
tracker.exportProgress();
```

CourseData Class

```
// Course data management  
const data = new CourseDataManager();  
  
// Methods  
data.loadCourse(courseId);  
data.getChapter(chapterId);  
data.getAllCourses();  
data.searchCourses(query);  
data.filterCoursesByLevel(level);
```

CourseCatalog Class

```
// Catalog functionality  
const catalog = new CourseCatalog();  
  
// Methods  
catalog.applyFilters();  
catalog.applySorting();  
catalog.renderCourses();  
catalog.getRecommendations(courseId);
```

Certificate System

Users can earn certificates by: 1. Completing 3 or more courses (configurable) 2. Achieving 90%+ overall completion rate

Check eligibility:

```
const certProgress = ProgressTracker.getCertificateProgress();
if (certProgress.eligible) {
  // Show certificate generation option
}
```

Security Considerations

- All data stored in client-side localStorage (no server)
- No authentication required (public courses)
- XSS protection via proper escaping
- Video URLs should use HTTPS
- Consider adding video access tokens for paid content

Future Enhancements

Planned Features: - ☐ Video quality switching implementation - ☐ Live streaming support - ☐ Interactive quizzes between chapters - ☐ Discussion/comments system - ☐ Download for offline viewing - ☐ Multi-language subtitle support - ☐ Picture-in-picture mode - ☐ Watch party feature - ☐ Mobile apps (iOS/Android) - ☐ Analytics dashboard

Backend Integration: - ☐ User accounts and authentication - ☐ Cloud progress sync - ☐ Course purchases/subscriptions - ☐ Instructor dashboard - ☐ Content management system - ☐ Video encoding pipeline - ☐ CDN integration

Browser Support

- Chrome 90+
- Firefox 88+
- Safari 14+
- Edge 90+
- Opera 76+
- Mobile Safari (iOS 14+)
- Chrome Mobile (Android 8+)

Troubleshooting

Video won't play: - Check video URL in course-data.js - Verify video format (MP4 recommended) - Check browser console for errors - Ensure CORS headers if using external CDN

Progress not saving: - Check localStorage quota (usually 5-10MB) - Verify browser allows localStorage - Check browser console for errors

Keyboard shortcuts not working: - Ensure focus is not in input/textarea - Check browser console for conflicts - Verify keyboard shortcuts modal shows correct keys

Styles not loading: - Verify path to player.css is correct - Check for CSS conflicts with parent site - Clear browser cache

License

This course player system is part of the HelixCode project and follows the same MIT License.

Contributing

To contribute to the course system:

1. Add new courses in course-data.js
2. Test across different devices
3. Update documentation
4. Submit improvements via pull request

Support

For issues or questions: - Check troubleshooting section above - Review browser console for errors - Open issue on GitHub - Contact HelixCode team

Built with **for the HelixCode community**

Last updated: January 2025